

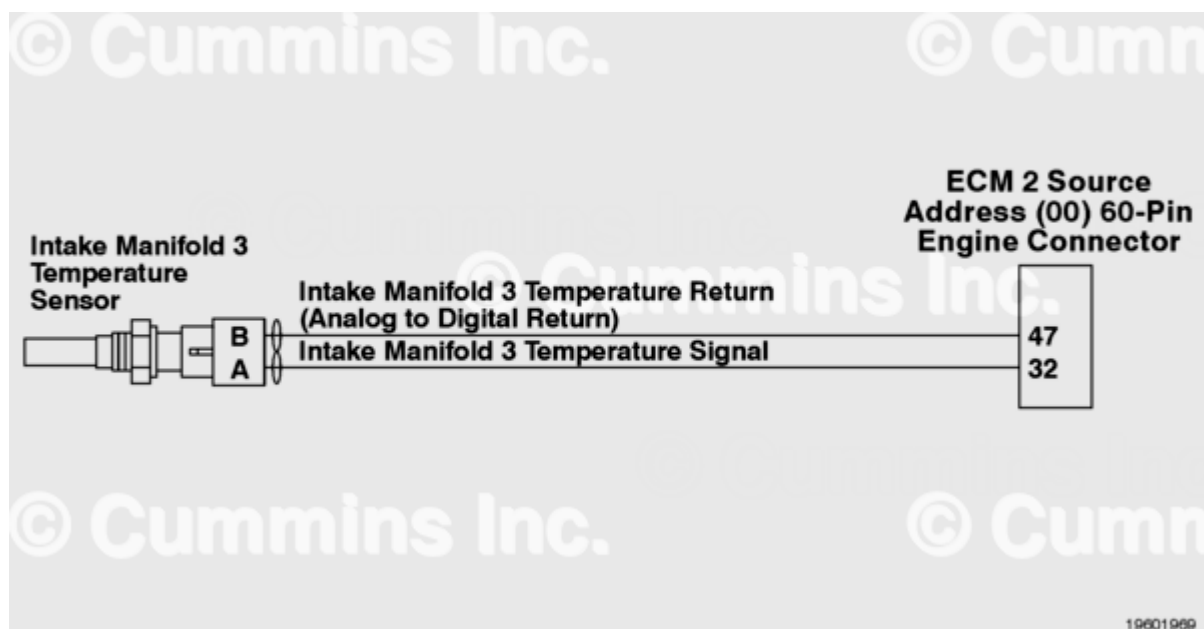
Fault Code: 161

Intake Manifold 3 Temperature Sensor Circuit - Voltage Below Normal or Shorted to Low Source

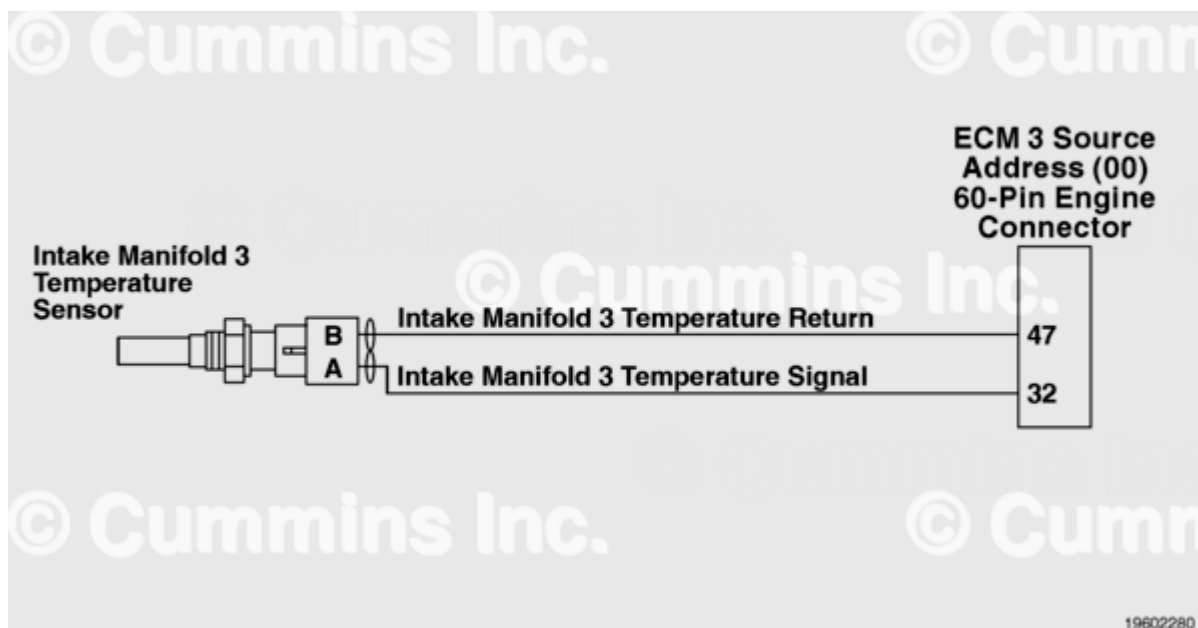
 Printable Version

Overview

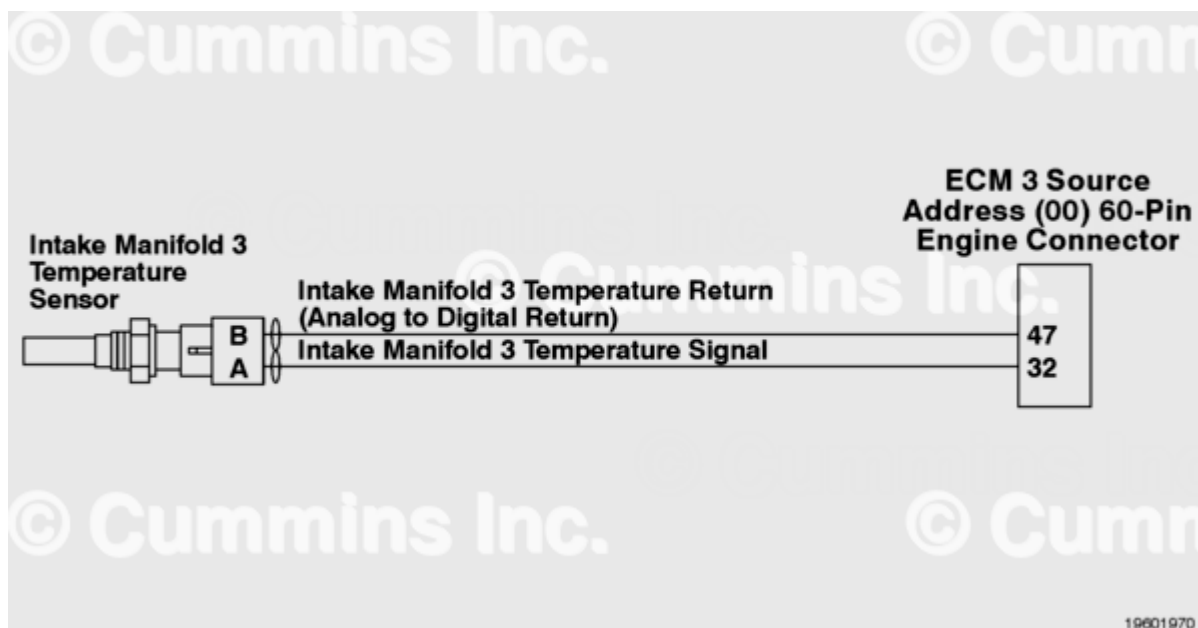
Codes	Reason	Effect
Fault Code: 161 PID(P): SPN: 1132 FMI: 4/4 Lamp: Amber SRT:	Intake Manifold 3 Temperature Sensor Circuit - Voltage Below Normal or Shorted to Low Source. Low signal voltage detected at intake manifold 3 temperature circuit.	Possible white smoke. Fan will stay ON if controlled by the ECM. No engine protection for intake manifold temperature.



QSK38 CM2150 Industrial - Intake Manifold 3 Temperature Sensor Circuit



QSK50 CM2150 Power Generation with Advanced Engine Monitoring - Intake Manifold 3 Temperature Sensor Circuit



QSK50 and QSK60 CM2150 Industrial/QSK50 and QSK60 CM2150 Marine/QSK60 CM2150 Drill Rig - Intake Manifold 3 Temperature Sensor Circuit

Circuit Description

The intake manifold 3 temperature sensor monitors intake manifold air temperature and passes information to the engine control modules (ECMs) through the engine harness.

Component Location

The intake manifold 3 temperature sensor is located in the left bank rear air intake manifold. Refer to Procedure 100-002 in Section E (</qs3/pubsys2/xml/en/procedures/28/28-100-002-tr.html>) of the K38, K50, QSK38, and QSK50 Service Manual, Bulletin 4021528, for a detailed component location view.

Shoptalk

There are multiple ECMs for the engine models included in this manual. Each ECM has an individual source address that displays when INSITE™ electronic service tool is connected. When troubleshooting a fault code, use the source address displayed in INSITE™ electronic service tool to determine which ECM and circuit is affected.

The intake manifold 3 temperature sensor shares return wires in the engine harness with other sensors. Opens and shorts in the engine harness can cause multiple fault codes to be active. Check fault codes with multiple counts first.

Possible causes of this fault code include:

- Signal circuit shorted to ground in the harness.
- Signal circuit shorted to return or ground in the sensor.

Temperature (°C)	Temperature [°F]	Resistance (ohms)
0	32	29k to 36k
25	77	9k to 11k
40	104	4.9k to 5.8k
100	212	600 to 700

Refer to Troubleshooting Fault Code 161. (</qs3/pubsys2/xml/en/procedures/122/122-t05-161.html>)